

AWS STRANDS SDK · STUDENT WORKSHOP

Building an AI Travel Booking Assistant

with Multi-Agent Orchestration

Hands-on demo · Live coding · Build your own agent



What We'll Cover

01



What is Strands SDK?

AWS's open-source framework for building AI agents

02



Live Demo

Travel booking assistant with real tools & agents

03



Under the Hood

Architecture deep-dive: tools, agents, orchestration

04



Your Turn!

Build & plug in your own agent using our template

What is AWS Strands SDK?

Strands is an open-source Python SDK from AWS that lets you build AI agents by defining tools and letting an LLM decide how to use them — automatically.

```
from strands import Agent, tool

@tool
def search_flights(origin, destination, date):
    """Search available flights"""
    return flights_api(origin, destination, date)

agent = Agent(tools=[search_flights])
agent("Find me flights NYC → Paris")
```



Tools

Python functions the AI can call



Agents

AI that reasons + picks tools



Orchestrator

Master agent that delegates

Travel Booking Multi-Agent System



How Strands Tools Work

1

User Message

User: "Book me a flight to Tokyo in June"

2

Agent Reasons

LLM decides: I need to call `search_flights()`

3

Tool Executes

Python function runs, real API called

4

Agent Responds

LLM sees result, generates final reply

YOUR CHALLENGE

Build Your Own Travel Agent!



1

Choose an idea

Budget estimator? Restaurant finder? Visa checker?

2

Fill in the template

Set agent name, system prompt & input fields

3

Define what it returns

Structured output: text, list, or score

4

Register your agent

One line to add it to the orchestrator



AGENT IDEAS

- 🍽️ Restaurant recommender
- 💰 Budget & cost calculator
- ☁️ Weather advisor
- 🛂 Visa requirements checker
- 💡 Language phrasebook
- 🚗 Local transport guide
- 📷 Top photo spots finder
- 🧳 Packing list generator

3 Things to Change — That's It!

```
# ✅ CHANGE 1: Give your agent a name
AGENT_NAME = "Restaurant Finder Agent"

# ✅ CHANGE 2: Write your system prompt
SYSTEM_PROMPT = """
You are a restaurant expert. Given a city
and cuisine type, suggest the best places.
"""

# ✅ CHANGE 3: Define inputs & what you return
INPUT_SCHEMA = {
    "city": "string",
    "cuisine": "string"
}
```

→ Orchestrator auto-discovers and uses your agent!

Your agent's identity — what should it be called?

Tell it how to behave — what is its expertise?

What info does it need from the user?

WRAP UP

What You Built Today



Used AWS Strands SDK

Open-source, production-ready agent framework



Built real AI tools

Python functions the LLM can invoke



Created a specialist agent

With custom system prompt & schema



Extended a live app

Plugged your agent into the orchestrator



Next Steps



Star Strands on GitHub



Add more tools to your agent



Deploy on AWS Bedrock



Read the Strands docs